Math 257 Definitions and Theorems

Week 2: Modules 5-8

Fall 2023

5 Linear Combinations

Definition 5.1. Consider $m \times n$-matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \text{ and } B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix}.$$

- We define their **sum** $A + B$ to be

$$A + B := \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{bmatrix}.$$

- If $c \in \mathbb{R}$ is a scalar, we define the ***scalar* product** cA to be

$$cA := \begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1n} \\ ca_{21} & ca_{22} & \cdots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{m2} & \cdots & ca_{mn} \end{bmatrix}.$$

Warning. Addition is only defined if A has the same number of columns and rows as B.

Definition 5.2. A **column vector** is an $m \times 1$-matrix. A **row vector** is a $1 \times n$-matrix.

Definition 5.3. If A is $m \times n$, the **transpose** of A is the $n \times m$ matrix, denoted by A^T, whose columns are formed from the corresponding rows of A. In terms of matrix elements: $(A^T)_{ij} = A_{ji}$.

Remark. The transpose of a column vector is a row vector and vice versa.

Definition 5.4. The **linear combination** of $m \times n$-matrices $A_1, A_2, \ldots, A_p$ with **coefficients** $c_1, c_2, \ldots, c_p$ is defined as:

$$c_1A_1 + c_2A_2 + \cdots + c_pA_p.$$

Definition 5.5. Given $m \times n$ matrices $A_1, \ldots, A_p$, we define their **Span**, $\text{Span}\,(A_1, \ldots, A_p)$, to be the set of all linear combinations of $A_1, \ldots, A_p$. That is,

$$\text{Span}\,(A_1, \ldots, A_p) := \{c_1A_1 + c_2A_2 + \cdots + c_pA_p \ : \ c_1, \ldots, c_p \text{ scalars}\}.$$

Definition 5.6. We denote the set of all column vectors of length m by $\mathbb{R}^m$.

Remark. Solving linear systems is the same as finding linear combinations!

Theorem 5.1. *Let $a_1, \ldots, a_n, b \in \mathbb{R}^m$. A vector equation*

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

has the same solution set as the linear system whose augmented matrix is

$$\left[\begin{array}{cccc|c} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{array}\right]$$

In particular, $\mathbf{b}$ can be generated by a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ if and only if there is a solution to the linear system corresponding to the augmented matrix.

6 Matrix-vector multiplication

Definition 6.1. Let $\mathbf{x}$ be a vector in $\mathbb{R}^n$ and $A = \begin{bmatrix} \mathbf{a}_1 & \ldots & \mathbf{a}_n \end{bmatrix}$ an $m \times n$-matrix. We define the product $A\mathbf{x}$ by

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \ldots + x_n\mathbf{a}_n.$$

Remark.

- $A\mathbf{x}$ is a linear combination of the columns of A using the entries in $\mathbf{x}$ as coefficients.
- $A\mathbf{x}$ is only defined if the number of entries of $\mathbf{x}$ is equal to the number of columns of A.

Theorem 6.1. *Let $A = \begin{bmatrix} \mathbf{a}_1 & \ldots & \mathbf{a}_n \end{bmatrix}$ be an $m \times n$-matrix and $\mathbf{b}$ in $\mathbb{R}^m$. Then the following are equivalent:*

- *$(x_1, x_2, \ldots, x_n)$ is a solution of the vector equation $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$*
- $\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ *is a solution to the matrix equation $A\mathbf{x} = \mathbf{b}$.*
- *$(x_1, x_2, \ldots, x_n)$ is a solution of the system with augmented matrix $\left[\begin{array}{c|c} A & \mathbf{b} \end{array}\right]$*

Notation. We will write $A\mathbf{x} = \mathbf{b}$ for the system of equations with augmented matrix $\left[\begin{array}{c|c} A & \mathbf{b} \end{array}\right]$.

7 Matrix Multiplication

Definition 7.1. Let A be an $m \times n$-matrix and let $B = \begin{bmatrix} \mathbf{b}_1 & \dots & \mathbf{b}_p \end{bmatrix}$ be an $n \times p$-matrix. We define

$$AB := [A\mathbf{b}_1 \quad A\mathbf{b}_2 \quad \cdots \quad A\mathbf{b}_p]$$

Theorem 7.1. *Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Then for every $\mathbf{x} \in \mathbb{R}^p$*

$$A(B\mathbf{x}) = (AB)\mathbf{x}.$$

Note. Other ways to compute AB.

- **Row-Column Rule** Let A be $m \times n$ and B be $n \times p$ such that

$$A = \begin{bmatrix} \mathbf{R}_1 \\ \vdots \\ \mathbf{R}_m \end{bmatrix}, \text{ and } B = \begin{bmatrix} \mathbf{C}_1 & \dots & \mathbf{C}_p \end{bmatrix}. \text{ Then}$$

$$AB = \begin{bmatrix} \mathbf{R}_1\mathbf{C}_1 & \dots & \mathbf{R}_1\mathbf{C}_p \\ \mathbf{R}_2\mathbf{C}_1 & \dots & \mathbf{R}_2\mathbf{C}_p \\ \mathbf{R}_m\mathbf{C}_1 & \dots & \mathbf{R}_m\mathbf{C}_p \end{bmatrix}$$

  and $(AB)_{ij} = \mathbf{R}_i\mathbf{C}_j = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$.

- **Outer Product Rule** Let A be $m \times n$ and B be $n \times p$ such that

$$A = \begin{bmatrix} \mathbf{C}_1 & \cdots & \mathbf{C}_n \end{bmatrix}, \text{ and } B = \begin{bmatrix} \mathbf{R}_1 \\ \vdots \\ \mathbf{R}_n \end{bmatrix}. \text{ Then}$$

$$AB = \mathbf{C}_1\mathbf{R}_1 + \cdots + \cdots + \mathbf{C}_n\mathbf{R}_n$$

8 Properties of Matrix Multiplication

Definition 8.1. The identity matrix I_n of size n is defined as

$$I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}.$$

Theorem 8.1. *Let A be an $m \times n$ matrix and let B and C be matrices for which the indicated sums and products are defined.*

(a) $A(BC) = (AB)C$ (associative law of multiplication)

(b) $A(B+C) = AB + AC$, $(B+C)A = BA + CA$ *(distributive laws)*

(c) $r(AB) = (rA)B = A(rB)$ *for every scalar* r*,*

(d) $A(rB + sC) = rAB + sAC$ *for every scalars* r, s *(linearity of matrix multiplication)*

(e) $I_m A = A = AI_n$ *(identity for matrix multiplication)*

Warning. Properties above are analogous to properties of real numbers. But **NOT ALL** properties of real number also hold for matrices. In particular, AB need not equal BA.

Theorem 8.2. *The transpose of a product is the product of transposes* ***in opposite order****:*

$$(AB)^T = B^T A^T$$

Definition 8.2. Let A be a square matrix. We write A^k for $A \cdots A$, k-times; that is A^k is obtained by multiplying A k-times with itself.